

1. The question in one paragraph

Is violence against civilians in a respondent's commune associated with how favorably they view the national leader, comparing people in the same region and survey wave? The outcome is a yes/no item answered "yes" by about 80% of respondents in every wave, so the association is expected to be modest. The analysis is built to estimate it carefully, with every model choice and decision rule fixed before the violence coefficient is examined.

Data structure. Three waves, each an independent two-stage sample: regions are strata, communes are the first stage, villages the second. About 140 communes per wave, many surveyed in more than one wave. ACLED events of violence against civilians are counted per commune and wave over the survey window. Only about 6% of the variation in violence is within communes across waves; the rest is between communes.

2. Notation

Symbol	Meaning
i, v, c, r, w	respondent, village, commune, region, wave
y_{icw}	1 if favorable view of the leader, 0 otherwise
E_{cw}	ACLED events in commune c , wave w
$V_{cw} = \log(1 + E_{cw})$	violence measure
$\mathbf{x}_i$	female, age per decade (centered), ethnicity, education, urban
u_{0c}, u_{1c}	commune random intercept and random violence slope
v_{rw}	region-by-wave random intercept
w_{vc}	village random intercept
τ	a random-effect variance

The violence measure

$V_{cw} = \log(1 + E_{cw})$. Zero events gives $V = 0$. A one-unit increase in V multiplies $(1 + E)$ by $e \approx 2.72$; an increase of $\log 2 \approx 0.69$ doubles $(1 + E)$. For large counts, doubling $(1 + E)$ is close to doubling E ; for small counts it is not (going from 0 to 1 event doubles $1 + E$). Because these steps are hard to picture, results are always also reported as predicted favorability at 0, 1, 5, and 20 events.

3. The model sequence

Every model is a logistic multilevel model fit with `lme4::glmer` by maximum likelihood (Laplace approximation), unweighted, on the same respondents. That makes log-likelihoods, AIC, and BIC comparable across M1 to M4.

M1. Empty model

$$\text{logit Pr}(y_{icw} = 1) = \gamma_{00} + u_{0c}, \quad u_{0c} \sim N(0, \tau_{00})$$

Reading it. γ_{00} is the log-odds of a favorable view in a typical commune; $\text{plogis}(\gamma_{00})$ is that as a probability. τ_{00} is how much communes differ. A logit model has no residual variance, so the intraclass correlation uses the variance of the standard logistic distribution:

$$\text{ICC} = \frac{\tau_{00}}{\tau_{00} + \pi^2/3}, \quad \pi^2/3 \approx 3.29$$

An ICC of 0.12 means 12% of the variation in the latent propensity to be favorable lies between communes. Any ICC well above zero justifies keeping a commune random intercept in every later model.

M2a and M2b. Violence, and does its effect vary by commune?

$$\text{M2a: } \text{logit Pr}(y_{icw} = 1) = \gamma_{00} + \gamma_{10} V_{cw} + u_{0c}$$

$$\text{M2b: } \text{logit Pr}(y_{icw} = 1) = \gamma_{00} + \gamma_{10} V_{cw} + u_{0c} + u_{1c} V_{cw}, \quad \begin{pmatrix} u_{0c} \\ u_{1c} \end{pmatrix} \sim \text{N}\left(\mathbf{0}, \begin{pmatrix} \tau_{00} & \tau_{01} \\ \tau_{01} & \tau_{11} \end{pmatrix}\right)$$

Reading it. γ_{10} is the violence effect in a typical commune. τ_{11} is how much that effect differs across communes. τ_{01} is the covariance between a commune's baseline favorability and its violence effect.

Test. $H_0 : \tau_{11} = 0$. $\text{LRT} = -2(\ell_{\text{M2a}} - \ell_{\text{M2b}})$. A variance cannot be negative, so the null sits on the boundary of the parameter space and the ordinary χ^2 test is wrong. M2b adds one variance and one covariance, so the p-value is a 50:50 mixture:

$$p = 0.5 \text{Pr}(\chi_1^2 > \text{LRT}) + 0.5 \text{Pr}(\chi_2^2 > \text{LRT})$$

Rule (fixed in advance). Keep the random slope only if $p < .05$ **and** M2b is not singular. If kept, the slope is carried into M3a, M3b, and M4.

Caution. A random slope for a commune-level exposure is identified only by within-commune change in violence across waves (about 6% of its variation). The test has little power. A non-significant result does **not** show the effect is the same in every commune; it shows the data cannot tell.

M3a. Region and wave dummies, covariates

$$\text{logit Pr}(y_{icw} = 1) = \gamma_{00} + \gamma_{10} V_{cw} + \sum_{r \neq \text{Bamako}} \alpha_r \text{Region}_r + \gamma_2 \text{Wave}_2 + \gamma_3 \text{Wave}_3 + \mathbf{x}_{i'} \boldsymbol{\beta} + u_{0c} [+ u_{1c} V_{cw}]$$

Bamako is the reference region; wave 1 is the reference wave. The bracketed term appears only if M2 kept the slope.

Reading it.

- γ_{10} : change in the log-odds of a favorable view per one-unit increase in V , for a typical commune, holding region, wave, and respondent characteristics constant. The violence effect now compares communes **within the same region and wave**.
- α_r : log-odds difference between region r and Bamako, other predictors held constant. Positive means more favorable than Bamako.
- γ_2, γ_3 : log-odds difference between wave 2 (or 3) and wave 1.
- The residual commune ICC, $\tau_{00}/(\tau_{00} + \pi^2/3)$, compared with M1, shows how much of the between-commune difference region, wave, violence, and composition explain.

Why region and wave are dummies, not random intercepts. Three waves are too few to estimate a variance. Regions are the sampling strata (every region is included, none is sampled). And a region random intercept assumes region effects are unrelated to violence, which fails here: northern regions have more violence and lower favorability, so between-region confounding would leak into γ_{10} (Bell & Jones, 2015).

M3b. Region-by-wave shocks

$$\text{logit Pr}(y_{icw} = 1) = \text{M3a} + v_{rw}, \quad v_{rw} \sim N(0, \tau_{rw})$$

The dummies stay. v_{rw} captures shifts that hit one region in one wave (a local event, a security operation) that separate region and wave dummies cannot absorb. With 24 region-wave cells, thin cells are pulled toward the main effects (partial pooling). Communes are nested in regions and crossed with waves.

Test. $H_0 : \tau_{rw} = 0$. M3b adds one variance, so the mixture is

$$p = 0.5 \Pr(\chi_0^2 > \text{LRT}) + 0.5 \Pr(\chi_1^2 > \text{LRT}) = \frac{1}{2} \Pr(\chi_1^2 > \text{LRT}),$$

half the usual χ_1^2 p-value (χ_0^2 is a point mass at zero).

Rule (fixed in advance). M3b proceeds only if $p < .05$, its AIC is lower than M3a's, and it is not singular. The winner is called M3.

Reading it. If M3b wins, favorability moves by region and wave in ways the separate dummies miss, and those shifts are now kept out of the violence coefficient. Compare γ_{10} in M3a and M3b: a large change means region-wave shocks were confounding the violence estimate. Expect τ_{rw} to be imprecise with 24 cells (in the simulation it was detected but estimated at 0.03 against a true 0.09).

M4. Final model: the design inside the multilevel model

$$\text{logit Pr}(y_{ivcw} = 1) = \text{M3} + \delta \log \tilde{w}_i + w_{vc}, \quad w_{vc} \sim N(0, \tau_v)$$

How each design feature enters:

Design feature	In M4
Strata (regions)	region dummies
First stage (communes)	commune random intercept u_{0c}
Second stage (villages)	village random intercept w_{vc}
Unequal selection (weights)	centered log weight $\log \tilde{w}_i$ as a covariate

glmer cannot use sampling weights (its `weights` argument means binomial trials or precision for these models), so the weight enters as a covariate, the standard model-based check for informative sampling (Pfeffermann et al., 1998; Gelman, 2007).

Reading it.

- γ_{10} in M4 is the final violence estimate: typical commune, village, and region-wave.
- τ_v : how much villages differ within communes.
- δ : if clearly nonzero, respondents with larger weights differ in favorability beyond the predictors, so selection is related to the outcome and the unweighted estimates may not represent the population. If near zero, the unweighted multilevel estimates are supported.

- Compare γ_{10} and its SE with M3: the ratio of standard errors is the variance cost of the design terms.

Design-based check (comparison only)

The same fixed effects as M3, fit with `survey::svyglm`: weights rescaled within wave, $w_i^* = w_i n_w / \sum_{i \in w} w_i$; strata are regions within wave; first stage communes within wave; second stage villages. Coefficients solve $\sum_i w_i^* \mathbf{x}_i (y_i - \mu_i) = 0$, with Taylor-linearized variance at the first stage (Binder, 1983):

$$\widehat{\text{Var}}(\hat{\beta}) = \hat{\mathbf{J}}^{-1} \left[\sum_h \frac{n_h}{n_h - 1} \sum_{j=1}^{n_h} (\mathbf{z}_{hj} - \bar{\mathbf{z}}_h)(\mathbf{z}_{hj} - \bar{\mathbf{z}}_h)' \right] \hat{\mathbf{J}}^{-1}, \quad \mathbf{z}_{hj} = \sum_{i \in (h,j)} w_i^* \mathbf{x}_i (y_i - \hat{\mu}_i)$$

Reading it. This coefficient is population-averaged, so it is smaller than the multilevel (typical-commune) coefficient by roughly

$$\beta_{\text{PA}} \approx \frac{\gamma_{10}}{\sqrt{1 + 0.346 \tau_{00}}}$$

(Zeger, Liang & Albert, 1988). Compare it with M4 by the z statistic, not the coefficient, and never by AIC or likelihood ratio tests (it has a pseudo-likelihood). If its z is much weaker than M4's, weighting and clustering matter and the result needs that caveat.

Sensitivity M3-MW. Usual level versus change

$$\text{logit Pr}(y_{icw} = 1) = \gamma_{00} + \gamma_W (V_{cw} - \bar{V}_c) + \gamma_B \bar{V}_c + \dots + u_{0c} + v_{rw}$$

$\bar{V}_c$ is the mean of V_{cw} over the waves in which commune c was surveyed (Mundlak, 1978).

Reading it. γ_B compares communes with persistently more violence to those with less (the “baseline” effect). γ_W asks whether a commune is less favorable in a wave when its violence is above its usual level (the “change” effect); it uses only repeat communes whose violence changed, so it is imprecise. If $\gamma_B \approx \gamma_W$, the single coefficient is a fair summary. Differencing from wave 1 was rejected: it drops every commune not surveyed in wave 1 and uses only the small within-commune variation.

4. Turning coefficients into numbers people understand

Odds ratios. $\exp(\gamma_{10})$ multiplies the odds of a favorable view each time $(1 + E)$ is multiplied by 2.72; $\exp(\gamma_{10} \log 2)$ each time $(1 + E)$ doubles. Example: $\gamma_{10} = -0.30$ gives $\exp(-0.30) = 0.74$ and $\exp(-0.30 \times 0.693) = 0.81$, a 19% drop in the odds per doubling.

Average marginal effect (AME). The average over respondents of $\partial \hat{\mu}_i / \partial V$, with random effects set to zero (re. form = NA). It is in probability units: an AME of -0.052 is -5.2 percentage points per one-unit increase in V .

Why the percentage-point effect is small. Near 80% favorable, the slope of the logistic curve is $p(1 - p) \approx 0.16$, against 0.25 at 50%. The same log-odds effect moves the percentage less when favorability is high.

Predicted favorability (the headline). Set every respondent's commune to E events, keep everything else as observed, average the predicted probabilities. Illustration from the simulation (not real results):

Events in commune	0	1	5	20
Predicted % favorable (M4)	82.3	79.1	73.2	65.4

5. What to check in each output

Output	Good sign	Warning sign
M1 ICC	clearly above 0	near 0: clustering is weak
Slope test	decision follows the rule	singular M2b: report as “not estimable”
Region-wave test	decision follows the rule	big change in γ_{10} between M3a and M3b
Log-weight δ	near 0	clearly nonzero: informative sampling
Design-based z	similar to M4	much weaker: weights or clustering matter
M3-MW	$\gamma_B \approx \gamma_W$	very different: baseline and change tell different stories
VIF	below about 2	high: predictors overlap
Binned residuals	intervals mostly cross 0, no pattern	a curve: missing nonlinearity
Convergence	lme4 messages are NULL	warnings: try another optimizer

Two software notes. `performance::check_convergence()` returns FALSE with lme4 2.0 because the derivatives it needs are no longer stored; use lme4’s own messages. `jtools::summ()` fails on confidence intervals for the intercept-only M1; use `confint = FALSE` there.

6. Plain-language summary for the client

Among people surveyed in the same region and survey round, those living in communes with more violence against civilians were less likely to view the leader favorably. If every commune had experienced no such events, about X% would be expected to hold a favorable view; with 5 events, about Y%. This mostly compares different communes, so it shows an association, not proof that violence caused the change.

Do not: call the result causal; show odds ratios to non-technical readers; compare `svyglm` with `glmer` by AIC or likelihood ratio tests; read one commune on the map; read a non-significant slope test as “the same effect everywhere”; add specifications after seeing results without labeling them exploratory.

7. Limitations to keep in mind

- **Mostly between communes.** Stable commune differences the covariates miss (remoteness, local history) could drive part of the estimate.

- **Unweighted multilevel models.** M4's log weight and the design-based check guard against informative sampling but do not replace weighting.
- **Violence measure.** Only surveyed communes have counts; ACLED undercounts in remote areas, most likely weakening the association.
- **Coverage.** Insecure areas are harder to survey; estimates describe the population that could be reached.
- **Design assumption.** Communes as the first stage and villages as the second should be confirmed with the survey team.

References

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